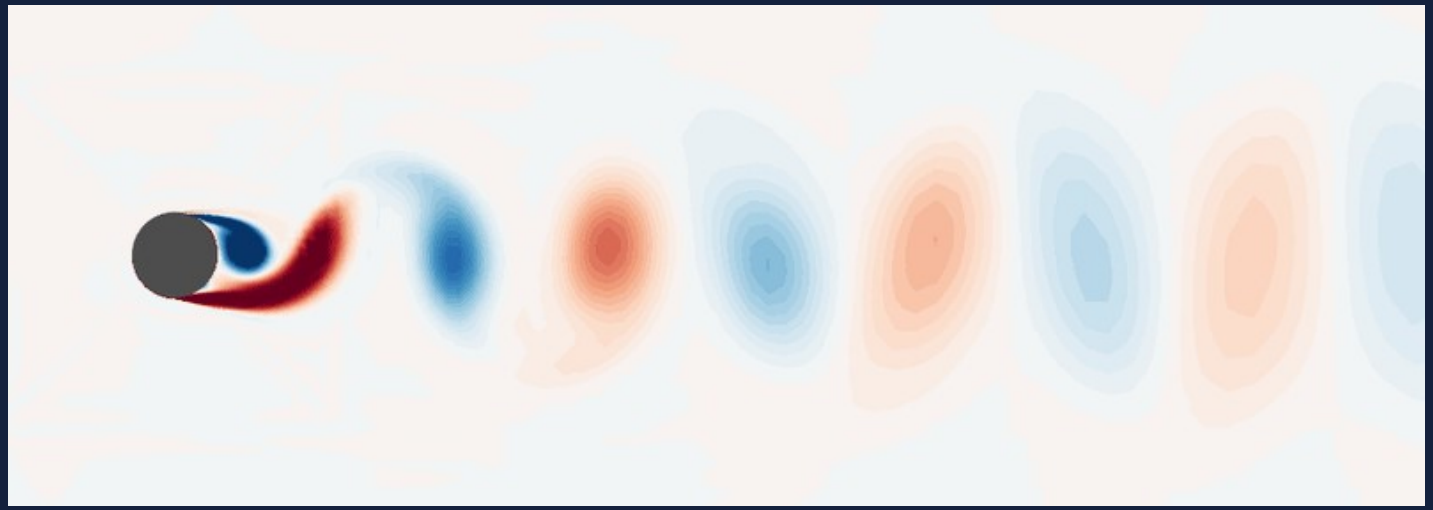
A COMPUTER EXPERIMENT

When wind meets a pole

What happens when air flows past a round object, and how we know the computer's answer can be trusted



Why wind around round things matters



It pushes

Wind pushes a pole, chimney or cable backwards. Engineers need that force to make it strong enough.



It shakes

Swirls of air break off on alternate sides and tug the object sideways, back and forth, at a steady rhythm.

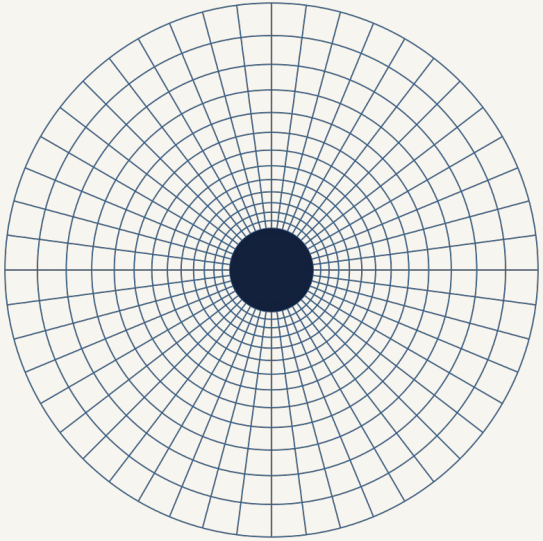


It can resonate

If that rhythm matches the structure's own natural swing, small tugs add up to large vibrations.

The experiment, inside a computer

- A round pole 10 cm wide, in a steady breeze of 10 m/s (36 km/h)
- The air around it is cut into up to 38,000 small cells
- In every cell, the laws of air motion are solved, 20,000 times for every second of flow
- Free, open-source software: OpenFOAM, on a laptop



Simplified sketch of the grid

What the computer showed: a trail of swirls



Red swirls spin one way, blue the other.

A pair, one top and one bottom, breaks off about 25 times a second.

What the wind does to every meter of pole

Push backwards

6 N

steady, about the weight of a 600 g bag of sugar hanging from each meter

Tug sideways

± 6.6 N

swinging smoothly back and forth, 25 full swings a second, one for each pair of swirls

For a 10 cm pole in a 10 m/s breeze. N = newton, the unit of force.

Can we trust the computer? Ask it three times

The same flow, solved on a coarse, a medium and a fine grid. If the answers stop changing as the cells shrink, the cells are small enough.

Grid	Cells	Swirls per second	Push (N per m)
Coarse	7,578	23.9	6.26
Medium	17,092	24.7	6.15
Fine	38,418	25.1	6.12

Medium to fine: every answer moved by less than 2 %. The same test with three time-step sizes agreed too.

THE VERDICT

The computer's own error is small, and we know how small

± 0.4 %

on the push

± 1.6 %

on the swirl rhythm

Accurate is not the same as true



Done: verification

The computer solves its equations correctly, with a known, small error.



Not yet: validation

Comparing with wind-tunnel measurements, to see if the equations describe real air, is future work.



Known simplifications

A flat slice instead of a full 3-D pole, and an approximate model of turbulence.

What comes next



Check against real air

Compare with wind-tunnel measurements, to learn how close the computer is to reality.



Test a real pole

Find a real pole's own swing rhythm, and the wind speeds where the swirls match it.



Let the pole move

Simulate a pole that can sway, to see how far it would actually shake.

Also: a full 3-D pole, other turbulence models, and more wind speeds.

What it is for



Safer structures. Engineers use numbers like these to size poles, chimneys and cables, and to keep their natural sway away from the rhythm of the swirls.



A habit worth having. Any simulation, of weather, cars or traffic, should be checked like this before its answer is trusted.

Check the computer first; then check it against reality.